RTSP/RTP Video Streaming Test Cases

Localhost

Server

Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : cpp.edu
   IPv6 Address. . . . . . . . . . . : 2620:df:8000:4703:0:2:c8e6:7a2f
   IPv6 Address. . . . . . . . . . . : fd36:77:2c7d:466b:c749:2c02:3629:8059
   Temporary IPv6 Address. . . . . . : fd36:77:2c7d:466b:d9c3:47eb:4a33:fb18
   Link-local IPv6 Address . . . . . : fe80::7950:7572:8bc9:fe13%9
   IPv4 Address. . . . . . . . . . . : 10.110.213.211
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : fe80::d6ad:71ff:fe6f:31cf%9
                                       10.110.208.1
```

Command:
python Server.py 8554

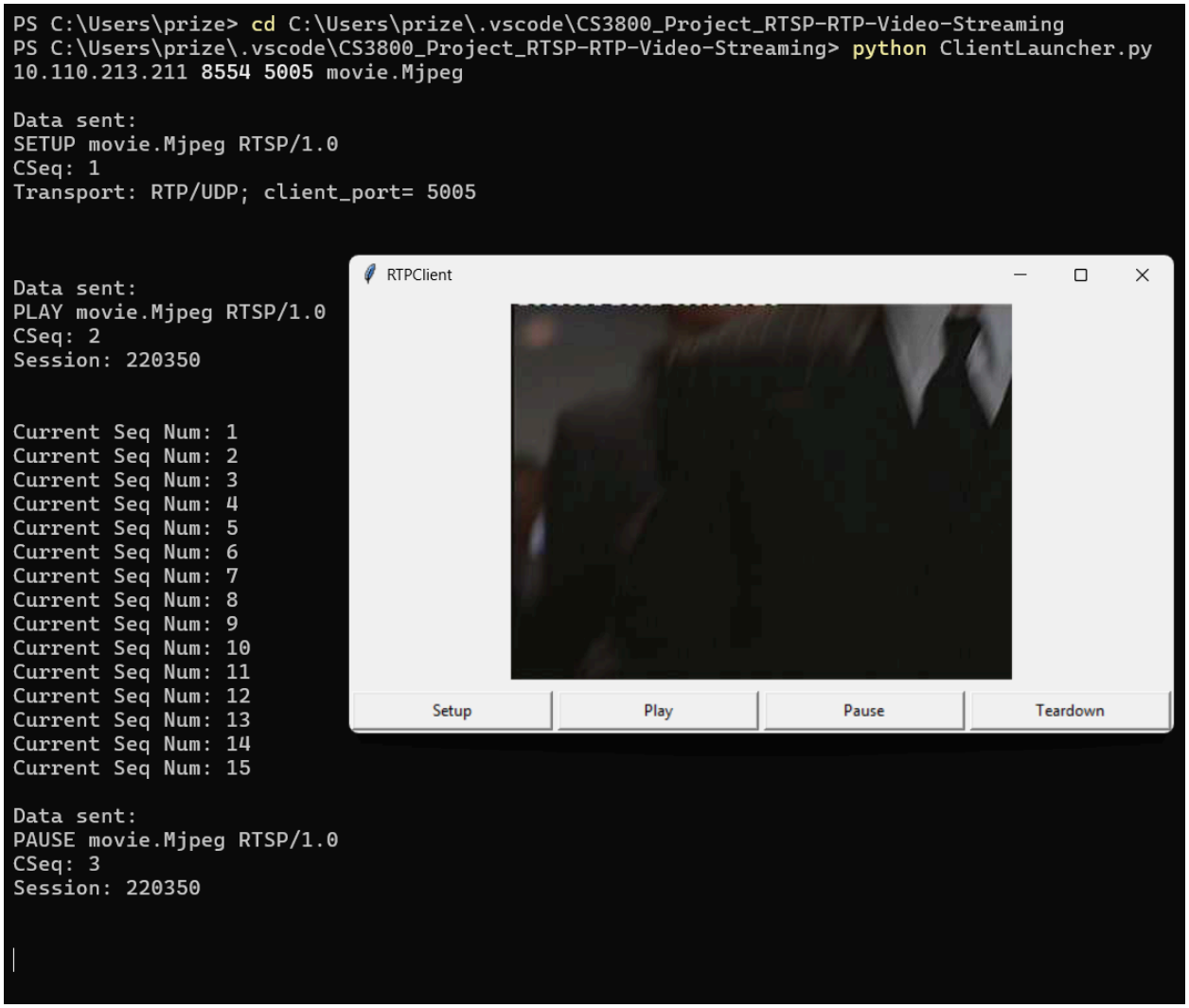
```
PS C:\Users\prize> cd C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming
PS C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming> python Server.py 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5005


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 220350


processing PLAY

Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Sending RTP to: 10.110.213.211 5005
Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 220350


processing PAUSE
```

Client

Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : cpp.edu
   IPv6 Address. . . . . . . . . . . : 2620:df:8000:4703:0:2:c8e6:7a2f
   IPv6 Address. . . . . . . . . . . : fd36:77:2c7d:466b:c749:2c02:3629:8059
   Temporary IPv6 Address. . . . . . : fd36:77:2c7d:466b:d9c3:47eb:4a33:fb18
   Link-local IPv6 Address . . . . . : fe80::7950:7572:8bc9:fe13%9
   IPv4 Address. . . . . . . . . . . : 10.110.213.211
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : fe80::d6ad:71ff:fe6f:31cf%9
                                       10.110.208.1
```

Command:
python ClientLauncher.py 10.110.213.211 8554 5005 movie.Mjpeg



Localhost x Kali Machine

Server on Localhost, Client on Kali

Server (LocalHost)

Command:
ipconfig

```
Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::be:40a8:619d:bc44%10
   IPv4 Address. . . . . . . . . . . : 192.168.91.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Command:

py -u “C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\Server.py” 8554

```
PS C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming> py -u “C:\Users\prize\
.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\Server.py” 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 162002


processing PLAY

Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 162002


processing PAUSE
```

Client (Kali)

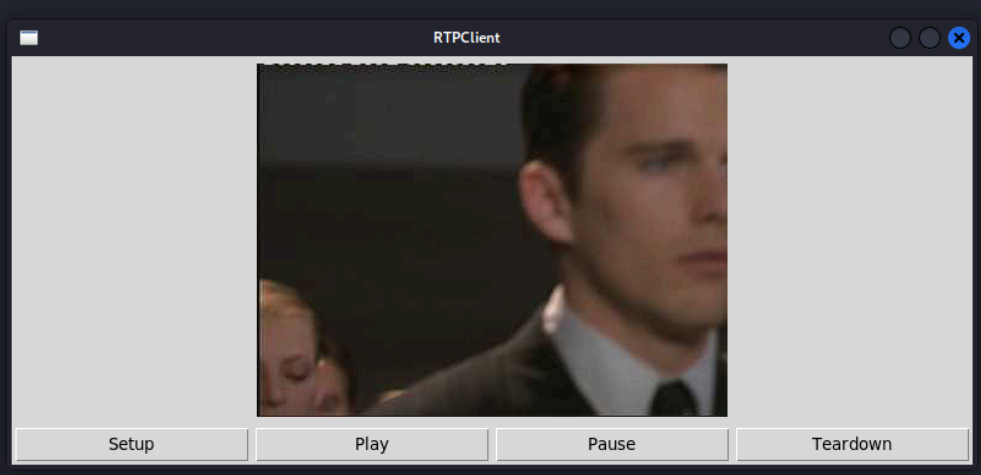
Command:

ip a

```
┌──(clarence㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 100
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 00:0c:29:a0:76:a2 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 00:0c:29:a0:76:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.91.128/24 brd 192.168.91.255 scope global dynamic noprefixroute eth1
       valid_lft 1379sec preferred_lft 1379sec
    inet6 fe80::20c:29ff:fea0:76ac/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Command:
python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/ClientLauncher.py" 192.168.91.1 8554 5000 movie.Mjpeg

```
┌──(clarence㉿kali)-[~]
└─$ python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/ClientLauncher.py" 192.168.91.1 8554 5000 mo
vie.Mjpeg

Data sent:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000



Data sent:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 162002


Current Seq Num: 1
Current Seq Num: 2
Current Seq Num: 3
Current Seq Num: 4
Current Seq Num: 5
Current Seq Num: 6
Current Seq Num: 7
Current Seq Num: 8
Current Seq Num: 9
Current Seq Num: 10
Current Seq Num: 11
Current Seq Num: 12
Current Seq Num: 13
Current Seq Num: 14
Current Seq Num: 15
Current Seq Num: 16
Current Seq Num: 17
Current Seq Num: 18
Current Seq Num: 19
Current Seq Num: 20
Current Seq Num: 21
Current Seq Num: 22
Current Seq Num: 23
Current Seq Num: 24
Current Seq Num: 25
Current Seq Num: 26
Current Seq Num: 27
```



Client on Localhost, Server on Kali

Server (Kali)

Command:
ip a

```
┌──(clarence㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
00
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 00:0c:29:a0:76:a2 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 00:0c:29:a0:76:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.91.128/24 brd 192.168.91.255 scope global dynamic noprefixroute eth1
       valid_lft 1796sec preferred_lft 1796sec
    inet6 fe80::20c:29ff:fea0:76ac/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Command:
python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/Server.py" 8554

```
┌──(clarence㉿kali)-[~]
└─$ cd /home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/

┌──(clarence㉿kali)-[~/CS3800_Project_RTSP-RTP-Video-Streaming]
└─$ python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/Server.py" 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 123739


processing PLAY

Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 123739


processing PAUSE

Data received:
TEARDOWN movie.Mjpeg RTSP/1.0
CSeq: 4
Session: 123739


processing TEARDOWN
```

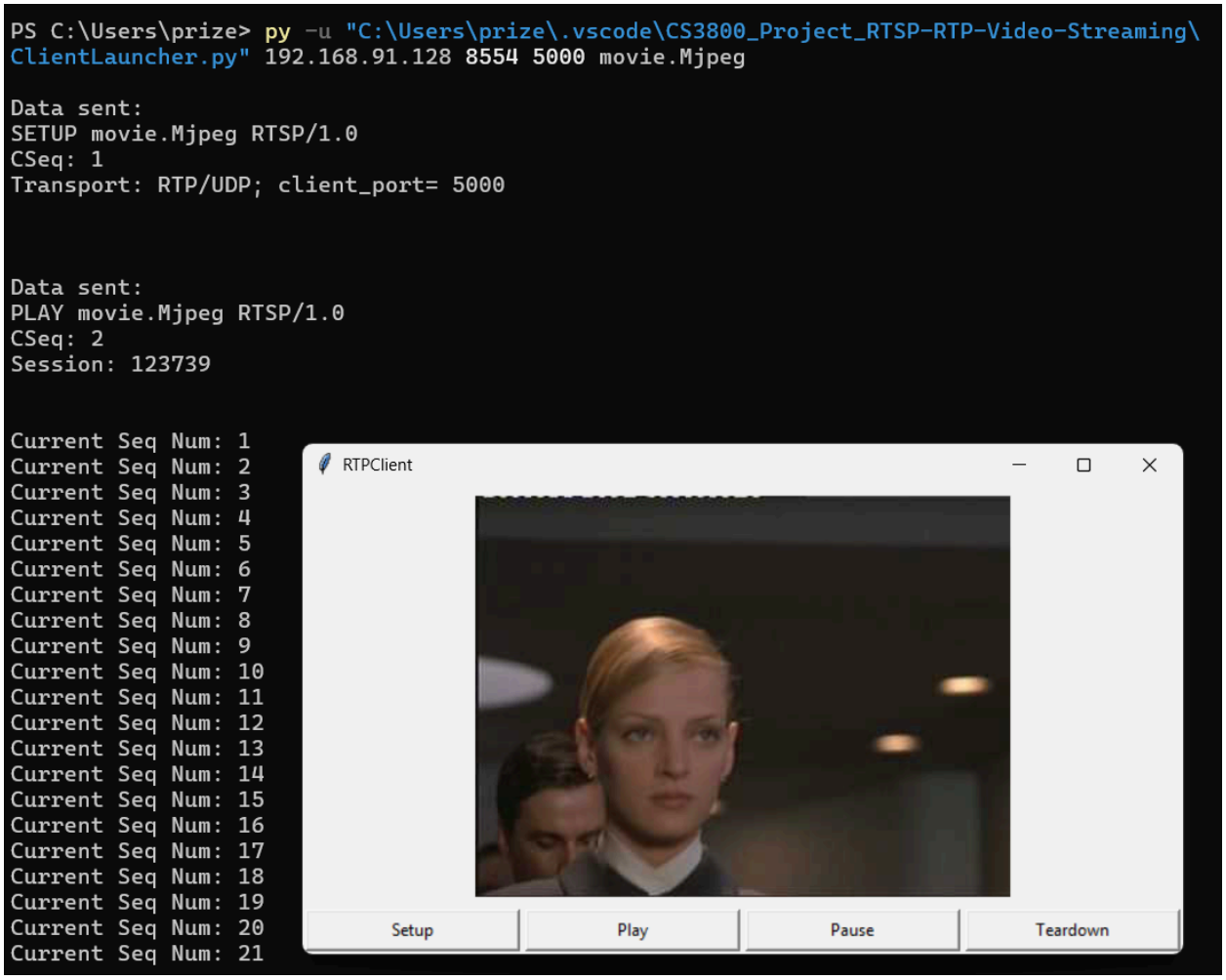
Client (Localhost)

Command:
ipconfig

```
Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::be:40a8:619d:bc44%10
   IPv4 Address. . . . . . . . . . . : 192.168.91.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Command:

py -u
"C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\ClientLauncher.py"
192.168.91.128 8554 5000 movie.Mjpeg



Kali Machine x Kali Machine

Server on Kali Machine 1

Command:
ip a

```
┌──(clarence㉿kali)-[~/CS3800_Project_RTSP-RTP-Video-Streaming]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 100
0
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
    link/ether 00:0c:29:3f:66:cf brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
qlen 1000
    link/ether 00:0c:29:3f:66:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.91.129/24 brd 192.168.91.255 scope global dynamic noprefixroute eth1
       valid_lft 1088sec preferred_lft 1088sec
    inet6 fe80::20c:29ff:fe3f:66d9/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Command:
python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/ClientLauncher.py" 192.168.91.129 8554 5001 movie.Mjpeg

```
┌──(clarence㉿kali)-[~]
└─$ cd /home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/

┌──(clarence㉿kali)-[~/CS3800_Project_RTSP-RTP-Video-Streaming]
└─$ python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/Server.py" 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 578065


processing PLAY

Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 578065


processing PAUSE

Data received:
TEARDOWN movie.Mjpeg RTSP/1.0
CSeq: 4
Session: 578065


processing TEARDOWN

Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5001


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 282871


processing PLAY
```

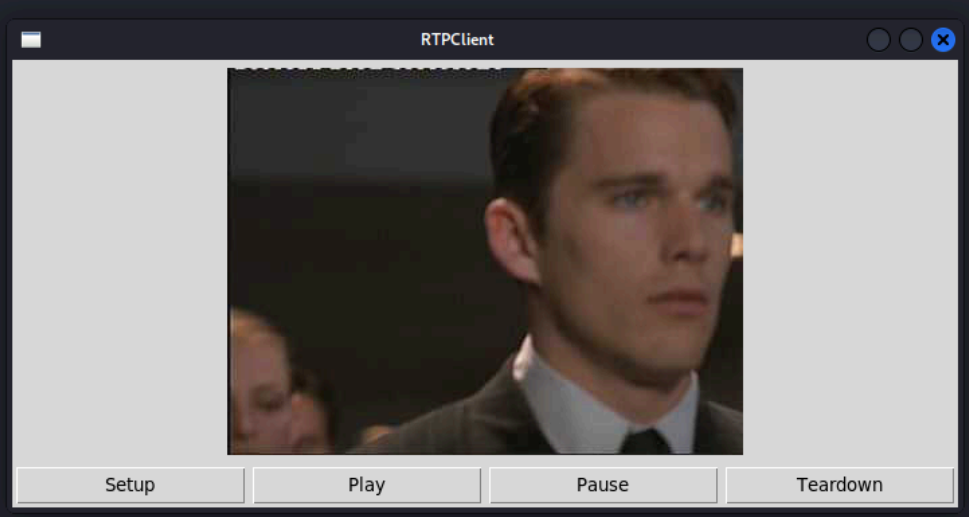
Client on Kali Machine 2

Command:
ip a

```
┌──(clarence㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
00
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 00:0c:29:a0:76:a2 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 00:0c:29:a0:76:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.91.128/24 brd 192.168.91.255 scope global dynamic noprefixroute eth1
       valid_lft 915sec preferred_lft 915sec
    inet6 fe80::20c:29ff:fea0:76ac/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Command:
python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/ClientLauncher.py"

```
┌──(clarence㉿kali)-[~]
└─$ python3 -u "/home/clarence/CS3800_Project_RTSP-RTP-Video-Streaming/ClientLauncher.py"
 192.168.91.129 8554 5001 movie.Mjpeg

Data sent:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5001



Data sent:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 282871


Current Seq Num: 1
Current Seq Num: 2
Current Seq Num: 3
Current Seq Num: 4
Current Seq Num: 5
Current Seq Num: 6
Current Seq Num: 7
Current Seq Num: 8
Current Seq Num: 9
Current Seq Num: 10
Current Seq Num: 11
Current Seq Num: 12
Current Seq Num: 13
Current Seq Num: 14
Current Seq Num: 15
Current Seq Num: 16
Current Seq Num: 17
Current Seq Num: 18
Current Seq Num: 19
Current Seq Num: 20
Current Seq Num: 21
Current Seq Num: 22
Current Seq Num: 23
Current Seq Num: 24
Current Seq Num: 25
Current Seq Num: 26
Current Seq Num: 27
```



Physical Machine x Physical Machine

Server on my Machine

Server (my machine)

Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7950:7572:8bc9:fe13%9
   IPv4 Address. . . . . . . . . . . : 10.186.116.68
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.186.116.50
```

Command:
py -u “C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\Server.py” 8554

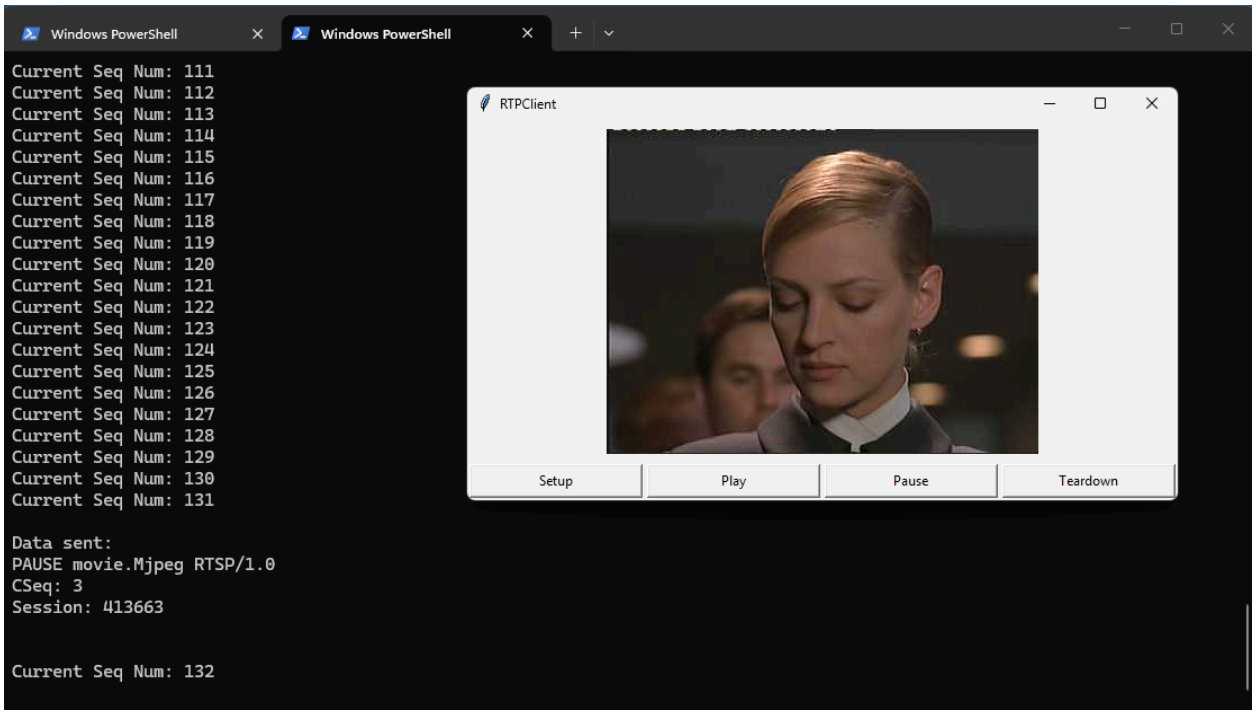
```
PS C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming> py -u "C:\Users\prize\
.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\Server.py" 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 413663


processing PLAY

Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 413663


processing PAUSE
```

Client (Sterling’s machine)

Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ea23:b56c:4be3:3c09%8
   IPv4 Address. . . . . . . . . . . : 10.186.116.75
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.186.116.50
```

Command:
py -u
“C:\Users\lordt\Downloads\CS3800_Project_Partner\CS3800_Project_Partner\ClientLauncher.py” 10.86.116.68 8554 5000 movie.Mjpeg



Client on my Machine

Server (Sterling’s machine)

Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ea23:b56c:4be3:3c09%8
   IPv4 Address. . . . . . . . . . . : 10.186.116.75
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.186.116.50
```

Command:
py -u “Server.py” 8554

```
PS C:\Users\lordt> cd C:\Users\lordt\Downloads\CS3800_Project_Partner\CS3800_Project_Partner
PS C:\Users\lordt\Downloads\CS3800_Project_Partner\CS3800_Project_Partner> py -u "Server.py" 8554
Data received:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000


processing SETUP

Data received:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 854227


processing PLAY

Data received:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 854227


processing PAUSE
```

Client (my machine)

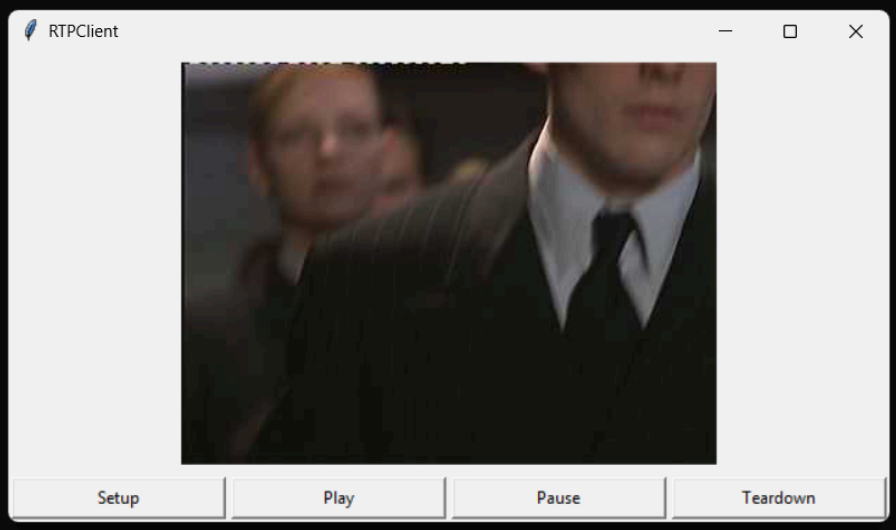
Command:
ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7950:7572:8bc9:fe13%9
   IPv4 Address. . . . . . . . . . . : 10.186.116.68
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.186.116.50
```

Command:

py -u “C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\ClientLauncher.py” sterling-ip 8554 5000 movie.Mjpeg

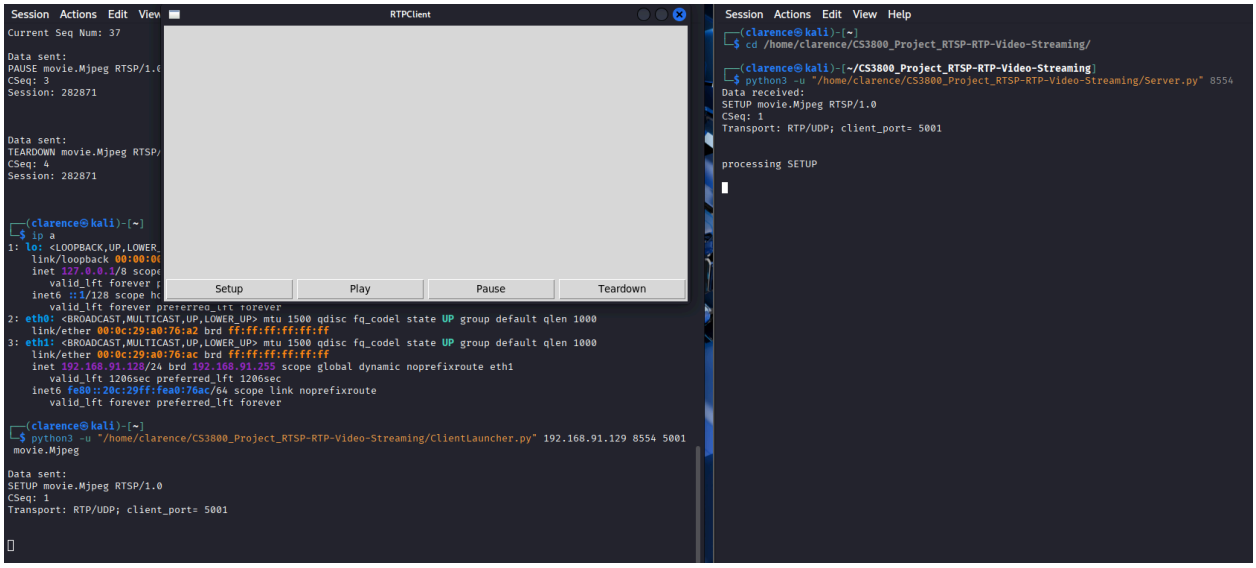
```
PS C:\Users\prize> py -u “C:\Users\prize\.vscode\CS3800_Project_RTSP-RTP-Video-Streaming\Cl
ientLauncher.py” 10.186.116.75 8554 5000 movie.Mjpeg

Data sent:
SETUP movie.Mjpeg RTSP/1.0
CSeq: 1
Transport: RTP/UDP; client_port= 5000



Data sent:
PLAY movie.Mjpeg RTSP/1.0
CSeq: 2
Session: 854227


Current Seq Num: 1
Current Seq Num: 2
Current Seq Num: 3
Current Seq Num: 4
Current Seq Num: 5
Current Seq Num: 6
Current Seq Num: 7
Current Seq Num: 8
Current Seq Num: 9
Current Seq Num: 10
Current Seq Num: 11
Current Seq Num: 12
Current Seq Num: 13
Current Seq Num: 14
Current Seq Num: 15
Current Seq Num: 16
Current Seq Num: 17
Current Seq Num: 18
Current Seq Num: 19

Data sent:
PAUSE movie.Mjpeg RTSP/1.0
CSeq: 3
Session: 854227
```



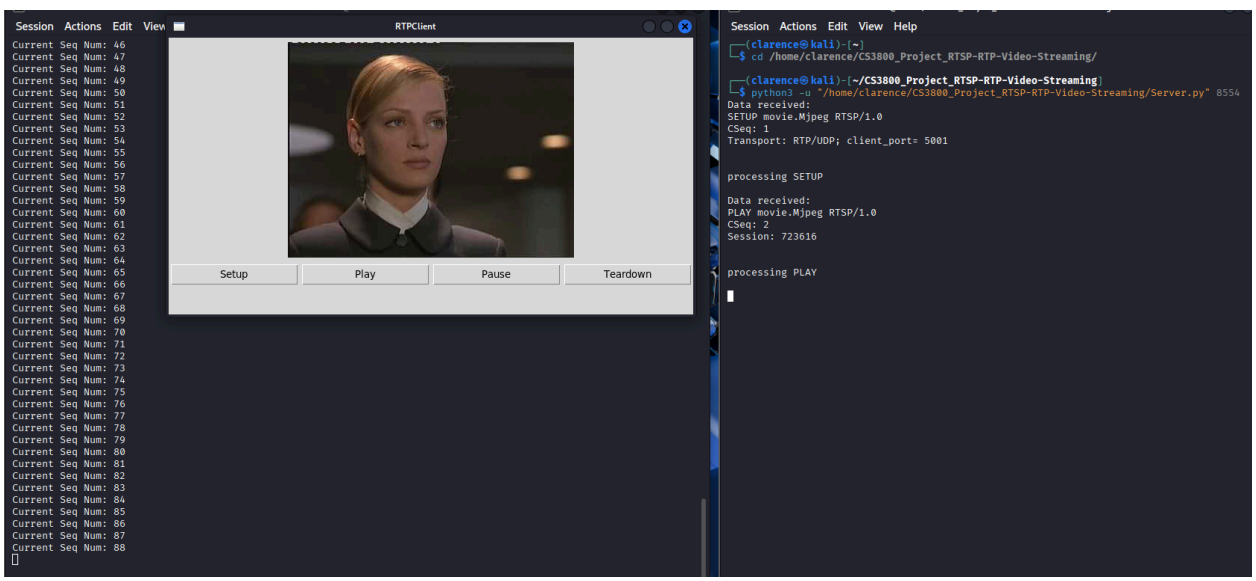
GUI Functionality

I will show the GUI functionality using two Kali machines.

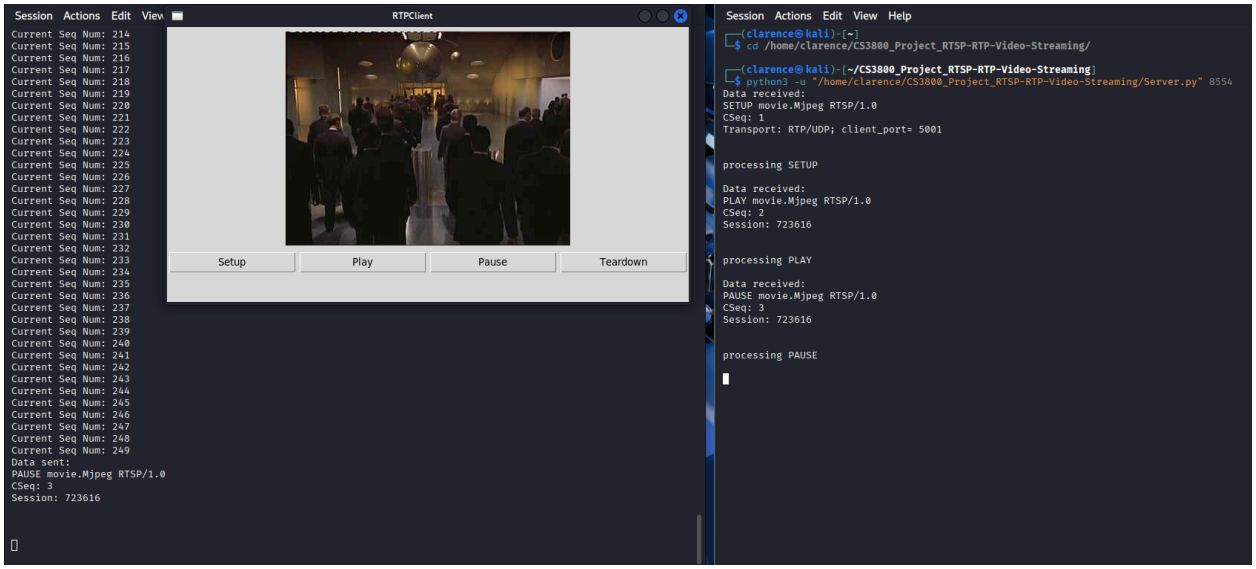
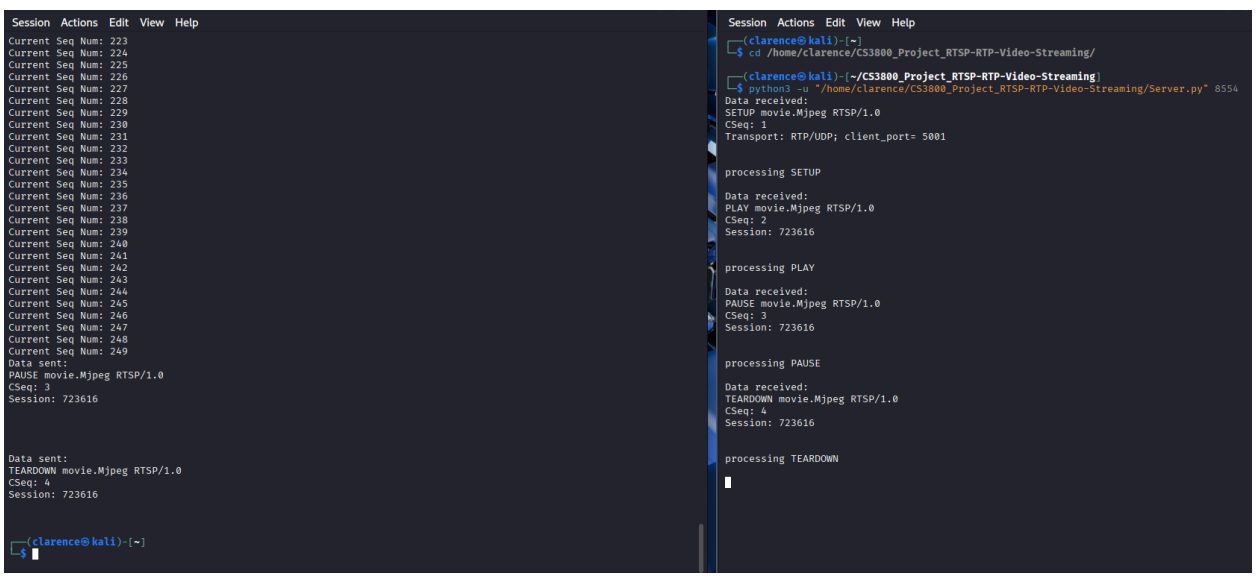
After Pressing the Setup Button



After Pressing the Play Button



After Pressing the Pause Button



After Pressing the Teardown Button